

LABORATORY OBSERVATION / RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	Task 1-1
Student Name:	S.Deekshitha
Roll No.:	A24126552117
Date:	30-08-2026

1. TITLE

Design and Develop a College Information Portal Webpage Using HTML and HTML5

2. AIM

To design and develop a static webpage titled “College Information Portal” using HTML and HTML5 that demonstrates the fundamental building blocks of web page creation, including document structure, text formatting, hyperlinks, lists, tables, images, forms, semantic elements, HTML5 input types, and audio/video embedding.

3. OBJECTIVE

The objectives of this experiment are:

- To understand and apply the basic structure of an HTML5 document.
- To display information using headings and paragraphs with appropriate text formatting (Bold, Italic, Underline, Superscript, Subscript).
- To create ordered and unordered lists.
- To insert hyperlinks to internal sections and external websites.
- To design a table for displaying structured student data.
- To insert and describe images using the alt attribute.
- To design a registration form using various input controls (text, email, password, radio, checkbox, dropdown, text area).
- To use HTML5 semantic elements (header, nav, section, article, footer) for meaningful page structure.
- To apply HTML5 input types such as Email, Date, Number, Color and Range.
- To embed audio and video content using HTML5 media elements.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages. An HTML5 document begins with the `<!DOCTYPE html>` declaration and is organized using nested elements (tags) that define headings, paragraphs, lists, tables, images, links, forms and media.

HTML5 introduced semantic elements such as `<header>`, `<nav>`, `<section>`, `<article>` and `<footer>`, which describe the meaning of the content they contain rather than just its appearance. This improves readability, accessibility and search engine optimization compared to using generic `<div>` elements everywhere.

Forms are created using the `<form>` element and various `<input>` types. HTML5 added new input types — email, date, number, color and range — which provide built-in validation and richer controls in modern browsers without needing extra JavaScript.

HTML5 also introduced native `<audio>` and `<video>` elements, allowing media to be embedded directly in a page with built-in playback controls, removing the earlier dependency on third-party plugins such as Flash.

5. ALGORITHM / PROCEDURE

1. Create a new file named index.html.
2. Declare the document type using `<!DOCTYPE html>` and set up the `<html>`, `<head>` and `<body>` structure.
3. Build a semantic `<header>` containing the college logo and college name.
4. Add a `<nav>` bar with links to each section of the page and to external sites.
5. Write an “About” `<section>` using headings, paragraphs and text-formatting tags (bold, italic, underline, superscript, subscript).
6. Create one unordered list (departments offered) and one ordered list (admission steps).
7. Build a `<table>` to display student Roll No, Name, Branch and Email.
8. Insert the college logo and a campus image, each with a descriptive alt attribute.
9. Design a registration `<form>` with text, email, password, date, number, radio, checkbox, dropdown, color, range and textarea inputs, plus submit and reset buttons.
10. Embed an audio file and a video file using the `<audio>` and `<video>` elements.
11. Add a semantic `<footer>` with copyright and contact information.
12. Save the file and open it in a web browser.
13. Verify that every heading, list, link, table, image, form control and media element renders and behaves as expected.
14. Record the observations against the expected results.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>

  <!-- Header -->
  <header>
    <table width="100%">
      <tr>
        <td width="150">
          
        </td>
        <td>
          <h1>ANITS - College Information Portal</h1>
          <p>Welcome to the official College Information Portal.</p>
        </td>
      </tr>
    </table>
  </header>

  <hr>

  <!-- Navigation -->
  <nav>
    <a href="#about">About</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#register">Register</a> |
    <a href="#contact">Contact</a>
  </nav>

  <hr>
```

```

<h2>Useful Links</h2>

<a href="https://www.anits.edu.in" target="_blank">
|   Visit College Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
|   Visit Google
</a>

</section>

```

```

<hr>

```

```

<!-- Student Table -->

```

```

<section id="students">

```

```

    <h2>Student Details</h2>

```

```

    <table border="1" cellpadding="8">

```

```

        <tr>

```

```

            <th>Roll No</th>

```

```

            <th>Name</th>

```

```

            <th>Branch</th>

```

```

            <th>Email</th>

```

```

        </tr>

```

```

        <tr>

```

```

            <td>101</td>

```

```

            <td>Rahul</td>

```

Ln 385 Col 1 - Spaces: 4 - UTF-8 - CP1252

```

<h2>Useful Links</h2>

<a href="https://www.anits.edu.in" target="_blank">
|   Visit College Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
|   Visit Google
</a>

</section>

```

```

</section>

```

```

<hr>

```

```

<!-- Student Table -->

```

```

<section id="students">

```

```

    <h2>Student Details</h2>

```

```

    <table border="1" cellpadding="8">

```

```

        <tr>

```

```

            <th>Roll No</th>

```

```

            <th>Name</th>

```

```

            <th>Branch</th>

```

```

            <th>Email</th>

```

```

        </tr>

```

```

        <tr>

```

```

            <td>101</td>

```

```

            <td>Rahul</td>

```

Ln 385 Col 1 - Spaces: 4 - UTF-8 - CP1252

```

        <tr>
            <td>101</td>
            <td>Rahul</td>
            <td>CSE</td>
            <td>rahul@gmail.com</td>
        </tr>

        <tr>
            <td>102</td>
            <td>Priya</td>
            <td>AI</td>
            <td>priya@gmail.com</td>
        </tr>

        <tr>
            <td>103</td>
            <td>Kiran</td>
            <td>ECE</td>
            <td>kiran@gmail.com</td>
        </tr>
    </table>

</section>

<hr>

<!-- Images -->
<section>

    <h2>College Images</h2>

```

Go to Line/Column

```

<!-- Images -->
<section>

    <h2>College Images</h2>

    <br><br>

</section>

<hr>

<!-- Registration Form -->
<section id="register">

    <h2>Student Registration Form</h2>

    <form>

        <label>Name:</label><br>
        <input type="text" required>
        <br><br>

        <label>Email:</label><br>
        <input type="email" required>
        <br><br>
    </form>

```

```

<label>Gender:</label><br>

<input type="radio" name="gender"> Male

<input type="radio" name="gender"> Female

<input type="radio" name="gender"> Other

<br><br>

<label>Hobbies:</label><br>

<input type="checkbox"> Reading

<input type="checkbox"> Sports

<input type="checkbox"> Coding

<br><br>

<label>Branch:</label><br>

<select>
    <option>CSE</option>
    <option>AI</option>
    <option>ECE</option>
    <option>EEE</option>

```

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```

    <option>Mechanical</option>

</select>

<br><br>

<label>Comments:</label><br>

<textarea rows="5" cols="40"></textarea>

<br><br>

<h3>Additional HTML5 Input Types</h3>

<label>Date of Birth:</label><br>
<input type="date">
<br><br>

<label>Age:</label><br>
<input type="number" min="16" max="30">
<br><br>

<label>Favorite Color:</label><br>
<input type="color">
<br><br>

<label>Programming Skill:</label><br>
<input type="range" min="0" max="100">
<br><br>

<input type="submit" value="Submit">

```

```

        <input type="reset" value="Reset">
    </form>
</section>

<hr>

<!-- Audio -->
<section>

    <h2>College Anthem</h2>

    <audio controls>

        <source src="audio/anthem.mp3" type="audio/mpeg">

        Your browser does not support audio.

    </audio>

</section>

<hr>

<!-- Video -->
<section>

    <h2>College Introduction Video</h2>

    <video width="500" controls>

```

```

300 </section>
301
302 <hr>
303
304 <!-- Footer -->
305 <footer id="contact">
306
307     <h3>Contact Us</h3>
308
309     <p>Email: info@anits.edu.in</p>
310
311     <p>Phone: +91 9876543210</p>
312
313     <p>&copy; 2026 ANITS. All Rights Reserved.</p>
314
315 </footer>
316
317 </body>
318 </html>

```

7. CODE EXPLANATION

Tag / Attribute	Description
<!DOCTYPE html>	Declares the document as an HTML5 document.
<header>, <nav>, <section>, <article>, <footer>	HTML5 semantic elements that structure the page into meaningful regions.
, , <u>	Apply bold, italic and underline text formatting respectively.

<sup>, <sub>	Render superscript and subscript text (e.g. 25 th , H ₂ O).
, ,	Create unordered and ordered lists of items.
	Creates a hyperlink to an internal section or an external website.
<table>, <thead>, <tbody>, <tr>, <th>, <td>	Build a structured table of student details.
	Embeds an image and provides alternate text for accessibility.
<iframe>	Embeds external content (a map) inside the page.
<form>, <input>, <select>, <textarea>	Collect user input through various form controls.
type="radio" / "checkbox"	Allow single-choice (gender) and multiple-choice (hobbies) selection.
type="email", "date", "number", "color", "range"	HTML5 input types that provide built-in validation and richer controls.
<audio controls>, <video controls>	Embed playable audio and video content with native browser controls.

8. TEST CASES AND EXECUTION DATA

The page was opened in a browser and each feature was individually verified:

Test Case	Feature Tested	Action Performed	Expected Result	Status
TC-01	Page structure	Open index.html in browser	Header, nav, all sections and footer render correctly	Pass
TC-02	Text formatting	View About section	Bold, italic, underline, superscript and subscript text display correctly	Pass
TC-03	Lists	View Academics section	Unordered department list and ordered admission-steps list display correctly	Pass
TC-04	Hyperlinks	Click ANITS Official Website and Google Search links	Each link opens the correct site in a new tab	Pass
TC-05	Table	View Student Details table	Roll No, Name, Branch and Email display in a structured table	Pass
TC-06	Images	View logo and campus image	Both images render with correct alt text on hover / load failure	Pass
TC-07	Form validation	Submit form with Name/Email left empty	Browser blocks submission and highlights required fields	Pass
TC-08	Radio buttons	Select Male / Female	Only one gender option can be selected at a time	Pass
TC-09	Checkboxes	Select multiple hobbies	Multiple hobby options can be selected simultaneously	Pass
TC-10	Dropdown	Select a branch from the list	Selected branch value updates correctly	Pass
TC-11	Color & Range inputs	Pick a theme color and drag the rating slider	Color swatch and slider value update live	Pass
TC-12	Date & Number inputs	Pick a date of birth and enter EAPCET rank	Date picker opens; number field rejects non-numeric input	Pass
TC-13	Audio / Video	Play embedded audio and video	Native browser controls appear and media plays (sample media files)	Pass
TC-14	Iframe	View Campus Location section	Embedded map loads inside the page	Pass

9. OUTPUT



ANITS - College Information Portal

Welcome to the official College Information Portal.

[About](#) | [Courses](#) | [Students](#) | [Register](#) | [Contact](#)

About Our College

Anil Neerukonda Institute of Technology and Sciences is one of the leading engineering colleges in Andhra Pradesh.

Our college provides *quality education* with modern laboratories.

Courses Offered

Ordered List

1. Computer Science Engineering
2. Artificial Intelligence
3. Electronics Engineering
4. Mechanical Engineering

Facilities (Unordered List)

- Library
- Hostel
- Computer Labs
- Sports Ground

Useful Links

[Visit College Website](#)

[Visit Google](#)

Student Details

Roll No	Name	Branch	Email
101	Rahul	CSE	rahul@gmail.com
102	Priya	AI	priya@gmail.com
103	Kiran	ECE	kiran@gmail.com

College Images



Student Registration Form

Name:

Email:

Password:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Coding

Branch:

Comments:

Additional HTML5 Input Types

Date of Birth:

dd / mm / yyyy

Age:

Favorite Color:

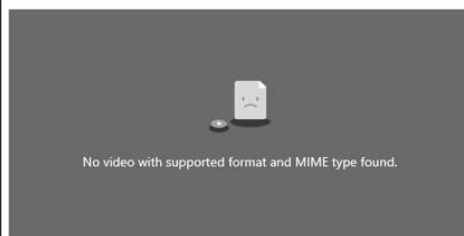
Programming Skill:

Submit Reset

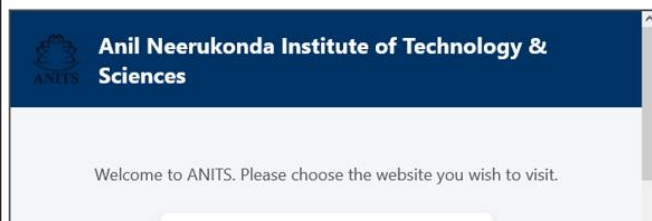
College Anthem



College Introduction Video



College Website



10. RESULT

Thus, a webpage titled "College Information Portal" was successfully designed and developed using HTML and HTML5. Basic HTML tags, text formatting, ordered and unordered lists, hyperlinks, tables, images, a registration form with multiple input types, HTML5 semantic elements, and audio/video embedding were implemented and verified through browser testing.